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ITA0406-R programming

R Programming - Activity 1

Integer Data Type

1. Write an R program to create an integer variable with value 100 and display its data type.

```
x <- 100L
print(x)
print(typeof(x))
print(class(x))
```

2. Write an R program to create two integer variables (25L and 15L) and find their sum, difference, and product.

```
a <- 25L
b <- 15L
sum_result <- a + b
diff_result <- a - b
prod_result <- a * b
cat("Sum:", sum_result, "\n")
cat("Difference:", diff_result, "\n")
cat("Product:", prod_result, "\n")
```

3. Write an R program to store the marks of 5 students as integers in a vector and calculate the average mark.

```
marks <- c(78L, 85L, 92L, 67L, 74L)
print(marks)
avg <- mean(marks)
cat("Average mark:", avg, "\n")
```

Numeric Data Type

4. Write an R program to create a numeric variable with value 45.75 and check its class.

```
num <- 45.75
print(num)
print(class(num))
```

5. Write an R program to calculate the area of a circle using a numeric radius value entered by the user.

```
radius <- as.numeric(readline(prompt = "Enter the radius: "))
area <- pi * radius^2
cat("Area of the circle:", area, "\n")
```

6. Write an R program to create a numeric vector containing five decimal values and find the maximum and minimum values.

```
values <- c(12.5, 8.75, 20.3, 5.6, 15.9)
print(values)
cat("Maximum value:", max(values), "\n")
cat("Minimum value:", min(values), "\n")
```

Complex Data Type

7. Write an R program to create a complex number and display its real and imaginary parts.

```
z <- complex(real = 3, imaginary = 4)
print(z)
cat("Real part:", Re(z), "\n")
cat("Imaginary part:", Im(z), "\n")
```

8. Write an R program to create two complex numbers and perform addition, subtraction, and multiplication operations.

```
z1 <- 2 + 3i
z2 <- 4 - 5i
print(z1 + z2)
print(z1 - z2)
print(z1 * z2)
```

Logical Data Type

9. Write an R program to compare two numbers and store the result in a logical variable. Display whether the first number is greater than the second.

```
num1 <- 25
num2 <- 18
result <- num1 > num2
print(result)
cat("Is first number greater than second?", result, "\n")
```

10. Write an R program to create two logical variables (TRUE and FALSE) and demonstrate the use of logical operators &, |, and !.

```
p <- TRUE
q <- FALSE
print(p & q)
print(p | q)
print(!p)
print(!q)
```